SLEEPING MACHINES • TUTORIAL 9

Forward Computation Teaches Credit

Causal reference learning, corrected gradients and sparse work budgets

Why this tutorial

A forward model produces more than a forecast: it exposes causal state, route effects and supervised observations that can teach a credit function. The useful question is which part of a future learning effect must still be estimated after reusing that structure. We derive a corrected route-policy gradient and separate its statistical and computational costs.

1 Issue the action before the observations

Let H denote information available when a route is chosen. The actor samples $A \sim \pi_\theta(\cdot \mid H)$, and a reference $b = b(H)$ is fixed independently of that sample. An exogenous recorded continuation U arrives afterward. Assume $U \perp A \mid H$: selecting an internal computational route does not change the data stream. This assumption fails when actions change the environment and its future observations.

Define $L(U, a)$ as the episode loss under route a, including subsequent causal parameter and optimizer updates. Credit concerns the contrast $D_a = L(U, a) - L(U, b)$, rather than only a frozen-model loss. Let $s_a = \nabla_\theta \log \pi_\theta(a \mid H)$ and $g_D(U) = \sum_a \pi_a D_a s_a$. Since $\sum_a \pi_a s_a = 0$, subtracting a reference independent of a leaves the policy gradient unchanged. This gradient covers route-policy parameters with the forward model snapshot fixed.

2 Use a predictor and repair its error

A learning-side predictor $h_a = h(H, U, a)$ can use already arrived U, although the actor could not use it when acting. Detach h from the current actor update. Write

$$g_h = \sum_a \pi_a h_a s_a, \qquad G = g_h + (D_A - h_A) s_A.$$

Conditioning on H, U gives

$$\mathbb{E}[G \mid H, U] = g_h + \sum_a \pi_a (D_a - h_a) s_a = g_D(U).$$

No prediction-accuracy assumption is needed for this mean identity. If a recorded positive-probability audit I reveals the contrast with probability p_a, use $I(D_A - h_A)s_A/p_A$ instead. Conditional audit selection must obey its declared propensity law.

Key idea

Future observations can teach credit after the action without leaking future information into inference. Route-dependent factual hidden state is not automatically a common exogenous observation. A valid issued reference and residual correction keep those distinctions explicit.

3 One reference learner carries shared adaptation

Suppose an addressed key write changes output logits while leaving a fixed deep carrier common to all routes. Let w_j^a be the readout state before target j in independently learning world a. Each target is scored before its own feedback update. Project route a through the reference learner's trajectory:

$$h_B(a) = \frac{1}{J} \sum_j [C_j(a, w_j^b) - C_j(b, w_j^b)].$$

The second term is the true reference loss. Exact cancellation yields

$$D_a - h_B(a) = \frac{1}{J} \sum_j [C_j(a, w_j^a) - C_j(a, w_j^b)].$$

The reference has learned from dense supervised observations; credit now needs only the route-induced difference between learner trajectories. Its projected alternative is not the actual independently learning alternative. That discrepancy is repaired by the factual/reference contrast.

Count actual work. One carrier may serve many route projections in this factorization. Every actual learner still owns its weights and optimizer moments. If the predictor already executes reference world b and factual world A executes normally, both losses are known. Set $p = 1$: discarding an audit cannot save a world already required by the predictor. Physically identical interventions may share one learner; distinct branches must not alias optimizer state.

4 Bound the remaining learner difference

Let z_j^a include weights and optimizer moments. Locally suppose the update map has state sensitivity α_j and route discrepancy $\beta_j(a, b)$ at the reference state. Then

$$\delta_{j+1} \le \alpha_j \delta_j + \beta_j, \quad \delta_0 = 0, \quad \delta_j \le \sum_{k<j} \beta_k \prod_{k<\ell<j} \alpha_\ell.$$

If target loss is L_j-Lipschitz in complete learner state,

$$|D_a - h_B(a)| \le \frac{1}{J} \sum_j L_j \sum_{k<j} \beta_k \prod_{k<\ell<j} \alpha_\ell.$$

This finite-episode bound separates route-induced update differences from optimizer amplification. Uniform $\alpha < 1$ gives a geometric bound; $\alpha = 1$ permits linear growth; $\alpha > 1$ permits expansion. Adam is not assumed globally contractive.

Key idea

A strong credit model can learn how interventions change learning, while forward/reference computation supplies the shared response. Cross-episode residual learning and general internal routes still require their own causal, information and cost tests.

5 Which noise remains?

The total covariance decomposes into

$$\mathrm{Cov}(G \mid H) = \mathbb{E}[\mathrm{Cov}(G \mid H, U) \mid H] + \mathrm{Cov}(g_D(U) \mid H).$$

Perfect realized credit removes action-sampling noise when g_h is evaluated exactly, but not variation of the true gradient between futures. Lower raw variance does not by itself guarantee better clipped or Adam updates.

6 Sparse queries have their own variance

Instead of summing every model route, draw Q independent candidates $C_q \sim \rho$ independently of A, conditional on H, U, with recorded positive support. Use

$$Z_C = \frac{\pi_C}{\rho_C} h_C s_C, \qquad G_Q = \frac{1}{Q} \sum_q Z_{C_q} + (D_A - h_A) s_A.$$

Its mean is still g_D, but its conditional covariance is

$$\mathrm{Cov}(G_Q \mid H, U) = \frac{\mathrm{Cov}(Z_C \mid H, U)}{Q} + \mathrm{Cov}((D_A - h_A) s_A \mid H, U).$$

With $h = D$, $\rho = \pi$ and $Q = 1$, ordinary one-sample policy noise returns. Cheap model queries can buy variance reduction, but discovery, normalization and score transport must be paid. If measured work is $C_0 + Qc$ and iid gradient variance is $V_0 + V_1/Q$, the continuous optimum of their product is $Q^* = \sqrt{C_0 V_1/(cV_0)}$. Online drift and correlations need additional analysis.

Exercises

Exercise 1. Why cannot an action-dependent reference be subtracted without changing the proof?

Solution. Its loss can vary with sampled a, so the discarded sum $\sum_a \pi_a L_b s_a$ need not be zero. A reference issued independently of A avoids this extra term.

Exercise 2. The predictor executes both learner worlds. Does reducing its audit probability to one quarter save reference work?

Solution. No. The reference loss is already paid and available; thinning discards information. Save that world only by redesigning how its trajectory is obtained, then audit the new approximation.

Exercise 3. Suppose future noise forms 99% of remaining variance. What can perfect residual prediction remove?

Solution. At most the remaining 1% in this decomposition, provided the model contraction stays exact. It cannot eliminate genuine future-gradient variation.

Where this appears in our work

The fixed-policy native diagnostic contracts the corrected mean against exact route-controller autograd. It reduces conditional action-gradient variance by99.92% versus prefix credit, with future variation making98.79% of remaining estimated total variance (12 TRAIN prefixes,16 futures). Carrier weights are fixed. The stochastic utility diagnostic is exploratory; fresh continuation confirmation is scored separately. See the forward-derived learner, reference-residual and sparse-model-term theory notes for full costs and scope.